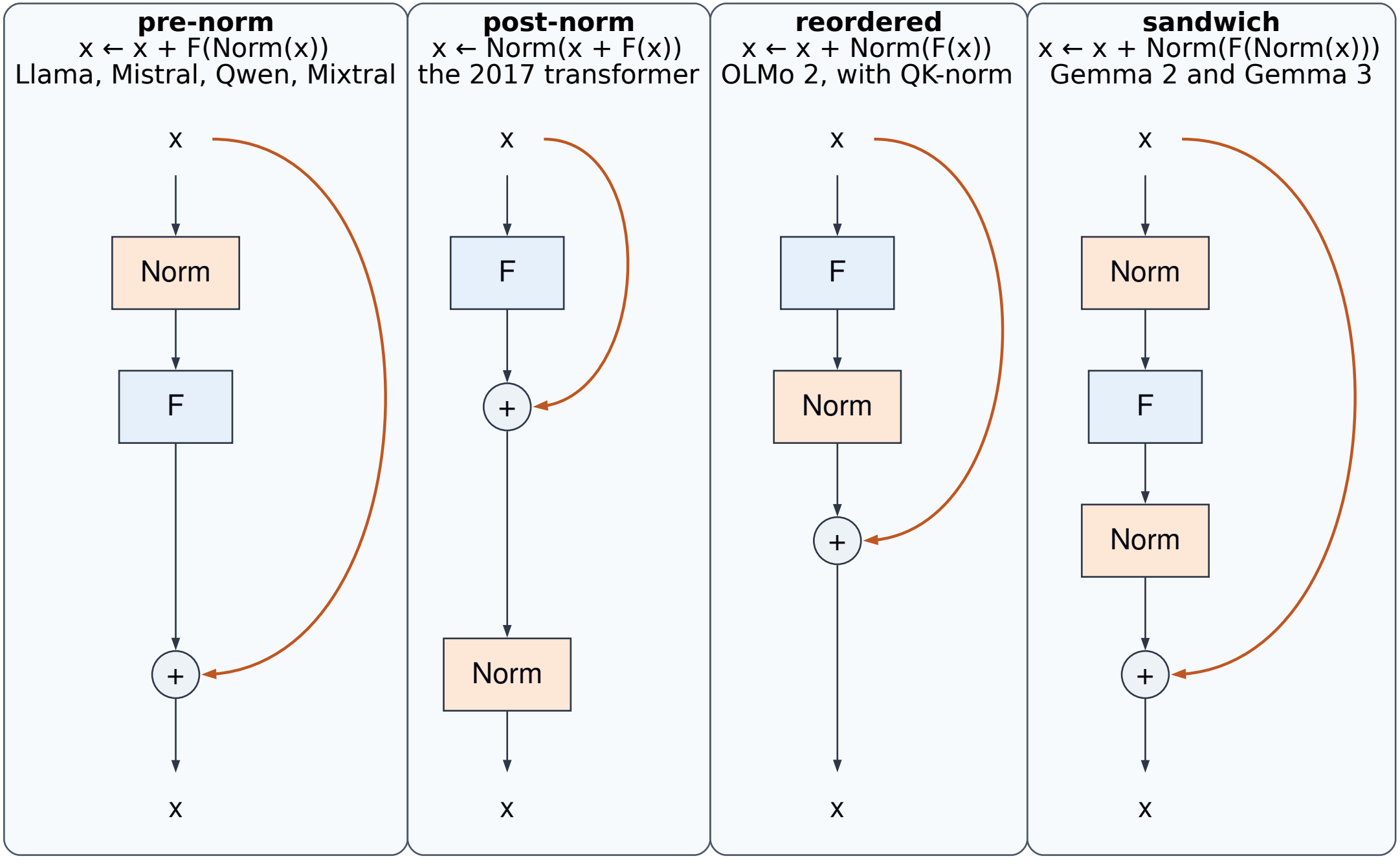
pre-norm
x ← x + F(Norm(x))
Llama, Mistral, Qwen, Mixtral
x
Norm
F
+
x
post-norm
x ← Norm(x + F(x))
the 2017 transformer
x
F
+
Norm
x
reordered
x ← x + Norm(F(x))
OLMo 2, with QK-norm
x
F
Norm
+
x
sandwich
x ← x + Norm(F(Norm(x)))
Gemma 2 and Gemma 3
x
Norm
F
Norm
+
x